Professional technical documentation:

The Compass

General Description

The Compass is a multi-level web platform designed to function as a counselor for pupils and students. The main goal of the site is to guide the user regarding the two most crucial decisions of their educational path: career selection and the strengthening of their linguistic skills.

Chapter 1: Home Page (index.html) - The Gateway to “The Compass”

1.1 What the user sees and what the page offers

Upon entering the home page (index.html), the first sight the user encounters is the **Hero Section**. Here, their gaze falls upon the central logo and a clear title that defines the identity of The Compass . The page offers immediate clarity: the user instantly understands that they are in a space dedicated to education and guidance.

Proceeding further, the user sees the **Info Sections** (Information Cards). These paragraphs are positioned to explain, in simple terms, the site's three pillars: finding courses, connecting with teachers, and the career orientation process. What the page offers at this point is **structure**: the user does not need to search; the information is presented to them in the order they need it.

Subsequently, the experience becomes more interactive with the appearance of the **Dynamic Circles** (Circle Cards). The user sees numbers increasing automatically (e.g., '100+ Students', '50+ Courses'). This visual effect is not merely decorative; it provides **Social Proof** and credibility, making the user feel secure about the quality of the services.

Finally, before reaching the end of the page, the user sees the invitation for the **Career Orientation Test**. This is a visually distinct section offering a solution to a problem: 'I don't know what to study? Here is a tool that will help me'. Overall, the home page acts as a 'roadmap' that transforms the user's uncertainty into a specific educational path.

1.2 How Navigation Works (The "Arrow" and the Menu)

Navigation on the home page is designed for maximum ease:

- **The Menu (Navbar):** It acts as a fixed reference point. Each link leads to a new subpage, allowing the user to switch from Online Courses to About Us with a single click.

- **The Back to Top Arrow:** In the bottom right corner of the screen, an arrow appears only when the user has scrolled down sufficiently. Upon clicking it, the page moves up smoothly (smooth scroll) to the top, facilitating an effortless return to the menu.

- **Theme Toggle:** A switch (slider) allows the user to change the entire appearance of the page from dark to light, offering comfortable reading depending on the environment's lighting conditions.

1.3 Main Page Code

"The 'brains' behind the home page reside in the index.html and script.js files.

- **HTML Structure:** It utilizes <header> for navigation and <main> or the content.

- **JavaScript (script.js) :** This houses the code for the Animated Counters. It leverages the .circle-card class and the IntersectionObserver API. This code 'watches' for when the user scrolls to the circles and then triggers the number animation, counting up from 0 to the target value.

- **CSS (style.css):** Defines the layout using Flexbox, ensuring the page remains aligned and visually appealing across all screen resolutions."

1.4 Interconnection with Other Pages

The home page functions as a traffic controller. Through <a> tags (hyperlinks) in the menu and buttons (e.g., Check Career Test), it directs the user to test.html for career orientation or Online-Courses.html for courses. All pages share the same style.css file and the same Theme Toggle mechanism (via localStorage), ensuring that if the user selects Light Mode on the home page, this setting automatically carries over to the rest of the pages.

Chapter 2: Courses and Instructor Profiles (Online-Courses.html & teacher-profile.html)

2.1 What the user sees and what the page offers

On the Online Courses page, the user comes into contact with the catalog of knowledge. The page is organized into thematic sections, such as Academic Pathways. What is offered here is specialization: the user does not simply see a list of courses, but rather sees Teacher Cards that include the photograph, the name, the title, and the university of the educator. This approach gives a human face to digital learning and allows the student to choose the mentor who inspires trust in them.

2.2 How navigation works (Sliders and Links)

The page utilizes an advanced method of navigation to avoid tiring the user with endless scrolling.

- The Sliders (Slider Track): Instead of the teacher cards occupying the entire length of the screen, they are arranged in horizontal rows. The user can move to the right and left by using the navigation arrows located at the edges of the slider. The movement is smooth and allows for a quick overview of multiple teachers within a minimal amount of space.

- Transition to Profile: Each card features the View Profile button. By clicking it, the user is directed to the teacher-profile.html page, where the information becomes more detailed.

- Interactive Calendar: Within the teacher's profile, the user sees a fully functional calendar. They can switch between months using the arrows and select a specific date for a lesson, which lights up (selected) to confirm their choice.

2.3 Main Page Code

The functionality of this section relies on three critical files:

- JavaScript (teachers.js): This acts as the data "reservoir." It contains all the elements regarding the teachers (biographies, photos, skills) stored in the form of objects.

- JavaScript (profile-loader.js): This code serves as the "transporter." When the user clicks "View Profile," the script reads the teacher ID from the URL and automatically "pulls" the correct information from teachers.js in order to build the page dynamically.

- CSS (profile-style.css): A specialized file that defines the aesthetics of the profiles, ensuring that the calendar and the cards for testimonials (student reviews) are displayed correctly and are user-friendly.

2.4 How pages are interconnected

The connection established here is smart and dynamic. The Online-Courses.html page does not send the user to multiple different pages, but rather to a single common template page (teacher-profile.html), by passing a parameter through the URL (e.g., ?id=1). The JavaScript code takes over and handles the rest:

1. It identifies which teacher the user has selected.
2. It changes the page title, the photograph, and the text in a fraction of a second.
3. It maintains the Theme (Light/Dark) that was selected on the home page, so that the experience remains consistent and professional.

2.5 User Experience in the Instructor Profile (Academics and Foreign Languages)

The profile page (teacher-profile.html) serves as the point where the user makes their final decision. The page offers a "showcase" that adapts according to the educator's specialty, providing the user with exactly the information they need.

A. What is seen in an Academic Instructor profile

For professors like Dr. Elena Papadopoulou (Ancient Greek), the user focuses on authority and expertise.

- Academic Background: The user sees the university and the degree titles, which offers them the assurance of profound knowledge.
- Skills: A list of specific fields is presented (e.g., Homeric Epics, Etymology), so that the user knows whether the teacher can cover their knowledge gaps.
- Teaching Philosophy: The user reads how the instructor approaches the subject, helping them understand if their chemistry will be good.

B. What is seen in a Foreign Language Instructor profile

In profiles like that of Isabelle Dubois (French), the experience shifts to focus on communication and academic credentials.

- Certifications: The user immediately sees whether the teacher handles preparation for specific diplomas (e.g., DELF, DELE).
- Multiculturalism: An insight is offered regarding the teacher's background and the languages they speak, which is critical for learning a foreign language.
- Levels: The user sees if the educator teaches at their specific proficiency level (e.g., A1 or B2), as determined by the Placement Test.

2.6 The Booking System (Booking & Calendar)

In every profile, regardless of the subject, the user encounters the Dynamic Calendar.

- Functionality: The user sees the current year and month. They can navigate through the upcoming months using the arrows. When they click on a date, it changes color (selected), confirming their choice.

- What it offers: It gives the user absolute control over their schedule. The page offers an immediate booking solution without the need to send emails or wait for a response regarding availability.

2.7 Technical Analysis: How code differentiates content

Even though the page appears visually identical, the code functions intelligently behind the scenes:

- profile-loader.js: This code reads the ID from the URL. If the ID belongs to an academic, it pulls authority-related data. If it belongs to a linguist, it pulls communication-related data.

- teachers.js: This is the database containing all the different fields (about, philosophy, skills) for each individual teacher.

- JavaScript DOM Manipulation: The code takes the text from the data file and automatically places it into the correct boxes on the HTML page, offering a lightning-fast and dynamic experience for the user.

Chapter 3: Interactive Tests and Evaluation (test.html & Language Quizzes)

3.1 What the user sees and what the page offers

When the user enters the Career Orientation Test page or any of the Language Placement Tests, they encounter a clean workspace free from distractions.

- In the Career Test: The user sees a series of questions regarding their preferences and way of thinking. The page offers self-awareness: at the end, instead of a simple score, they receive a comprehensive analysis of their personality type along with recommendations for academic departments and professions.

- In the Language Tests: The user sees questions of escalating difficulty (from A1 to B2). The page offers placement: it tells the user precisely which level they are at, so they know exactly which class they should enroll in.

3.2 How the process works (Progress Bar and Results)

The functionality of the tests is designed to keep the user engaged:

- Progress Bar: At the top of the screen, a dynamic bar fills up as the user answers questions. This provides visual confirmation of how close they are to completion, which helps reduce fatigue.

- Dynamic Question Switching: The user does not see all the questions at once. By clicking the "Next" button, the current question disappears with a smooth effect and the next one appears.

- Instant Results: As soon as the test is completed, the screen changes automatically to display the results using charts or lists, without requiring a page reload.

3.3 Main Test Code

- JavaScript (test.js): It utilizes the RIASEC algorithm. Each answer provided by the user adds points to one of the six categories (Realistic, Investigative, Artistic, etc.). At the end, the showResults() function compares the scores and displays the two most dominant types.

- JavaScript (main-script.js): It manages the language tests. It uses a large data object (allQuizzesData) where the questions are divided by level. The script checks in real-time whether an answer is correct and calculates the final proficiency level (e.g., "Your level is B1").

- CSS (language-style.css): It defines the colors for correct and incorrect answers (green/red) and ensures that the test environment is user-friendly and legible on mobile devices.

3.4 How pages and code are interconnected

The interconnection of the tests with the rest of the site is strategic:

- From the Test to the Courses: At the end of each test, a "Check Online Courses" button appears. This is not accidental. The code directs the user from the result (e.g., "Computer Science suits you") to the corresponding courses page, closing the service cycle.

- Shared Logic: Although the Career Test and the Language Tests are different, they share the same coding philosophy (DOM Manipulation). They use JavaScript to "write" content to the screen dynamically, without the user feeling like they are changing sites.

- Theme Persistence: As with the other pages, the Dark/Light mode selection remains constant, ensuring that the user completes the test in an environment that they themselves have customized.

Chapter 4: Team Identity and Technical Architecture (About-Us.html & File System)

4.1 What the user sees on the "About Us"

On the About Us page, the user moves from the functional aspect to the human element. The page offers the "why" behind "The Compass."

- Visual Content: The user sees texts describing the vision and values of the platform, often accompanied by images that symbolize guidance and progress.

- Consistency: Even though the content changes, the user feels "at home," as the page maintains the same Header and Footer as the home page, reinforcing the professional image of the brand.

4.2 How the "Theme Flash Prevention" System works

One of the most important technical features that the user encounters across all pages (including the About Us page) is the avoidance of the "white flash."

- The Functionality: When the user has selected Dark Mode, the system must be aware of this preference even before the page is displayed.
- The Arrow and Navigation: Just like on the home page, the "Back to Top" arrow is present to facilitate the reading of long texts, allowing the user to return to the menu instantaneously.

4.3 Main Code and Architecture (Back-end Logic)

Here we explain how the files running in the background work together:

- HTML (About-Us.html): It uses the same template as the other pages for the sake of uniformity. In the <head> section, it includes a critical script (IIFE) that reads the localStorage immediately.

- CSS (style.css): It functions as the "central source of truth" for the design. All HTML files are linked to it, ensuring that if you change a color in the CSS, it will change across the entire site simultaneously.

- JavaScript (Shared Logic): Although each page has its own scripts (e.g., test.js or profile-loader.js), they all rely on common JavaScript principles for managing the DOM (Document Object Model).

4.4 How all files are interconnected (Site Network)

The interconnection of "The Compass" resembles a well-organized network:

1. Data Interconnection: The teachers.js file feeds information to both the Online-Courses.html page (for the cards) and the teacher-profile.html page (for the details).

2. Design Interconnection: The profile-style.css and language-style.css files do not function autonomously, but rather build upon the core variables of style.css.
3. Experience Interconnection: Through the use of URL Parameters (e.g., id=1), the site "transfers" information from page to page without requiring a database, making it extremely fast and lightweight.

Chapter 5: Technical Implementation and Code Analysis

In this chapter, the architecture of the application, the technologies selected, and the business logic of the individual subsystems of "The Compass" are analyzed.

5.1 Tech Stack Overview

For the development of the platform, a Client-Side architecture was adopted, aiming for execution speed and the minimization of required resources.

- HTML5 (Structure): Semantic tags (<header>, <main>, <article>) were utilized to ensure the optimal structure of the document, enhancing SEO and accessibility.

- CSS3 (Formatting): The user interface (UI) design was implemented based on modern Flexbox and Grid standards. For the management of the color palette, central variables (CSS Variables in :root) were created, allowing for universal and immediate theme switching (Dark/Light mode) without the need to modify multiple files.

- JavaScript ES6+ (Logic): Modular code was developed without the use of external libraries (Vanilla JS), ensuring full control over performance and memory management.

5.2 Subsystem 1: Vocational Orientation Test Mechanism

The functionality of the Orientation Test (test.html) is based on data processing according to the RIASEC model. The relevant logic is located in the test.js file.

A. Data Structure The questions are not statically included in the HTML code but are stored in an array of objects (const questions). Each object contains the text of the question and the personality type it represents: { text: "...", type: 'R' } (where R = Realistic).

B. Scoring Algorithm A scores object is used to record the points per category: let scores = { R: 0, I: 0, A: 0, S: 0, E: 0, C: 0 };

Upon form submission, the code executes the following steps:

1. Iterate: All of the user's answers are traversed.
2. Accumulate: For each positive answer, the counter for the corresponding type is increased (e.g., scores[question.type]++).
3. Sort: The score object is converted into an array and a descending order sort is applied to identify the highest values.
4. Result Generation: The two dominant types (Primary & Secondary) are selected and the corresponding result texts are dynamically injected into the DOM.

5.3 Subsystem 2: Language Quiz Engine

The management of language tests (foreignlanguages.html) is handled through a central script (main-script.js), which controls the examination flow.

A. State Management

Monitoring the test flow is achieved through state variables

1. `currentQuestionIndex`: The index of the current question within the data array.
2. `score`: The current total of correct answers.

B. Initialization

Through the `initializeQuiz` function, the dataset to be loaded is defined based on the selected language (e.g., `allQuizzesData.french`). This function:

1. Clears the content of the question card.
2. Dynamically creates buttons (`<button>`) for each possible answer.

3. Adds **Event Listeners** to the buttons so that the correctness check (`checkAnswer`) is executed upon clicking.

C. Level Calculation

Upon completion of the test, the algorithm calculates the success rate. Based on predefined **Thresholds**, the user is categorized into the appropriate proficiency level (A1, A2, B1, B2), and the corresponding result card is displayed.

5.4 Subsystem 3: Dynamic Instructor Profiles

To avoid code redundancy and enhance scalability, a dynamic page generation system was developed (**profile-loader.js**) that retrieves data from the **teachers.js** file.

A. URL Parameter Handling : Instructor identification is performed via the URL Query String (e.g., `teacher-profile.html?id=1`). The ID is retrieved using the **URLSearchParams API**.

B. Data Injection: The `find()` method locates the corresponding object in the `teachers` array. Data is then placed into HTML elements via `textContent` (for text), `src` (for images), and `forEach` loops (for skills and testimonials):

- `textContent`: Used for the name, title, and biography.
- `src`: Used for the profile picture.
- `forEach Loop`: Used for generating the skills list and testimonials.

C. Availability System (Calendar Logic) A calendar algorithm was integrated into the profile, which dynamically calculates the days of the current month using the logic: `new Date(year, month, 0).getDate()`. This functionality enables the accurate display of dates and interactive appointment selection, without any dependence on an external server.

5.5 Directory Structure and File Organization

To ensure maintainability, the application files were organized into a hierarchical structure. Separating multimedia, styles, and scripts allows for easy navigation through the code.

The root directory structure is configured as follows:

- **Root /:** Contains the core HTML files (`index.html`, `About-Us.html`, `Online-Courses.html`, `test.html`).
- **/css:** A directory containing style sheets (`style.css`, `profile-style.css`, `language-style.css`), separated by functional unit.
- **/js:** A directory containing logic files (`script.js`, `main-script.js`, `teachers.js`, `profile-loader.js`, `test.js`).
- **/Images:** The central folder for all multimedia assets.
    - **/Images/Teacher-Faces:** A subfolder containing teacher photographs optimized for rapid loading.
    - **/Images/Icons:** Vector files (SVG/PNG) for logos and UI icons.

Chapter 6: Design Approach and User Experience (UI/UX)

6.1 Color Palette and Color Psychology

The selection of colors in "The Compass" is not accidental but is based on the psychology of color within an educational context.

- **Primary Blue (#3b82f6):** Blue was chosen as the dominant color because it symbolizes trust, knowledge, and tranquility. In an educational platform, it is critical for the user to feel a sense of security.

- **Dark Mode Background (#121828):** For the dark mode, a shade of "Navy Blue" was selected instead of absolute black. This reduces eye strain and offers a more sophisticated aesthetic.

- **Feedback Colors:**
    - **Green (#28a745):** Used exclusively for rewards (correct answer, successful booking).
    - **Red (#dc3545):** Used sparingly for error indication to avoid discouraging the student.

6.2 Typography

For the website's typography, a combination of the **Jost** and **Inter** fonts was selected (via Google Fonts).

- **Jost:** Used for headings and logos. It is a geometric sans-serif typeface that provides a modern and friendly character.

- **Inter:** Used for the body text. It was chosen for its exceptional readability on small screens and digital environments, ensuring that long texts (e.g., teacher biographies) are easy to read.

6.3 Mobile-First Design and Responsiveness

Given that the target audience (pupils/students) primarily uses mobile devices, the design was implemented using **Responsive Design** principles. Through **CSS Media Queries**, the layout adapts automatically:

- **Course Grids:** The layout transitions from 3 columns on desktops to a single column on mobile devices (`grid-template-columns: 1fr`).

- **Navigation Menu:** The menu scales to touch-friendly sizes, ensuring ease of use on smaller screens.

- **Graphics and Progress Bars:** These elements scale dynamically without losing their clarity or resolution.

Chapter 7: Quality Assurance and Performance Optimization

7.1 Testing Strategy

Prior to final delivery, the platform underwent exhaustive testing to ensure proper functionality.

1. **Cross-Browser Testing:** The website was tested across the most popular browsers (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari) to ensure that CSS Styles and JavaScript Animations are rendered consistently.
2. **Functional Testing:** User simulations were conducted to verify critical functions:

    a. **Verification of the Career Test Algorithm:** Confirmed that the RIASEC algorithm generates distinct and accurate personality profiles based on varying user input.
    b. **Calendar Validation:** Verification that the dynamic calendar prevents the selection of past dates through visual indicators (e.g., grayed-out dates) and logic blocks.

c. **Theme Toggle Persistence:** Testing to ensure that the user's Dark/Light mode preference remains stored in `localStorage` even after a page refresh (F5) or browser restart.
a.

7.2 Performance and Lighthouse Score

Google Lighthouse was used to measure performance. The choice of a **Pure Client-Side** architecture (without heavy frameworks) resulted in exceptionally high scores:

- **Performance: >95/100** – Achieved due to the absence of unnecessary third-party libraries and optimized asset delivery.
- **Accessibility: >90/100** – Attained through the deliberate use of high-contrast colors and Semantic HTML5 elements.
- **SEO:** Optimized via unique `<title>` and `<meta>` tags for every subpage, ensuring better visibility in search engine results.

Chapter 8: Maintenance and Scalability Manual

This chapter serves as a technical guide for the management and expansion of the platform, analyzing the procedures for adding new content.

8.1 Procedure for Adding a New Instructor

The platform's architecture allows for the addition of new instructors without the need to create new HTML files. This process is carried out exclusively through the **teachers.js**.

Steps:

1. Open the **js/teachers.js** file.
2. Locate the **const teachers** array.
3. Add a new object with the following structure:

JavaScript

```
{
 id: [NEW_UNIQUE_NUMBER],
 name: "Teacher Name",
 title: "Title/Specialty",
 university: "Graduating University",
```

```
 image: "Images/Teacher-Faces/new-photo.jpg",
 contact: { email: "email@example.com", phone: "..." },
skills: ["Skill 1", "Skill 2"],
about: "Biography text...",
philosophy: "Educational philosophy...",
 testimonials: [{ text: "Review...", author: "Student" }]
}
```

4. The platform automatically recognizes the new entry and dynamically generates the profile page.

8.2 Language Quiz Addition Process

To introduce a new language (e.g., Chinese), the following actions are required in the **main-script.js** file:

1. **Extend the `allQuizzesData` object:** Add a new key (e.g., `chinese`) to the central data structure.

2. **Input the question array:** Format the data using the structure `{ question: "...", answers: [...], correct: "..." }`.

3. **Create a new HTML file:** Generate a copy of an existing file (e.g., `french.html`), rename it (e.g., `chinese.html`), and update the call ID in the `initializeQuiz` function to match your new key.

8.3 Security and Data Management

Since the application is **Client-Side**, no sensitive personal data is maintained on a server.

- **Local Storage:** Used exclusively for storing theme preferences (Dark/Light mode). No names, emails, or test results are stored, ensuring user **privacy** and alignment with **GDPR Compliance**.

- **Input Sanitization:** Within contact forms and tests, the code prevents the execution of malicious scripts (**XSS Protection**). This is achieved by handling text via `textContent` instead of `innerHTML`, ensuring that any input is treated as literal text rather than executable code.

Chapter 9: Project Evaluation (Conclusion)

The completion of "The Compass" platform marks the achievement of a dual goal: providing high-quality educational counseling through a technologically sound digital environment.

9.1 Technical Excellence Assessment

The choice of a **"Pure Client-Side"** architecture (HTML/CSS/Vanilla JS) proved to be strategically sound. It offered:

- **Maximum Speed:** The absence of heavy libraries (such as React or Angular) ensured nearly instantaneous loading times.
- **Zero Maintenance Cost:** The application does not require complex servers or databases, making it both sustainable and economical.
- **Stability:** The simplicity of the code minimizes the potential for bugs and ensures uninterrupted operation.

9.2 Educational Impact

Beyond the code, "The Compass" constitutes a tool with a social impact. Through the RIASEC algorithms and Language Tests, it democratizes access to vocational orientation. It provides every student, regardless of location or financial means, with a free, reliable, and scientifically documented self-awareness tool.

In summary, this project is not merely a website, but a comprehensive digital ecosystem that connects knowledge with need, technology with education, and the student with their future. Its modular structure and clean code guarantee that it will continue to serve its purpose with reliability and efficiency for years to come.